

2(a)(i)	time taken for, indigo carmine / solution / indicator, to, turn blue / change colour ;	1
2(a)(ii)	<i>any one from:</i> number / mass, of beads (of immobilised cells) ; size / surface area, of beads ; concentration of alginate ; density / number, of cells / protoctists ; ambient / background / AW, light (intensity) ; pH ; concentration of indicator ; species / type, of protoctist / unicell / AW ;	1
2(b)(i)	2.55 ;	1
2(b)(ii)	<i>any three from:</i> 1 <i>idea that</i> increasing, hydrogencarbonate / carbon dioxide, (concentration) increases, the rate of photosynthesis / the rate / photosynthesis / the rate of reaction / ora ; 2 <i>idea that</i> increasing, light intensity, increases, the rate of photosynthesis / the rate / photosynthesis / the rate of reaction / ora ; 3 ref. to when carbon dioxide / hydrogencarbonate, is a limiting factor using information from Table 2.1 ; 4 ref. to when light is a limiting factor using information from Table 2.1 ;	3

Question	Answer	Marks
2(b)(iii)	any two from: 1 <i>idea of</i>, (small) range / number / AW, of different concentrations of, (sodium hydrogencarbonate / carbon dioxide) solution ; 2 <i>idea of</i>, (small) range / number / AW, of different light intensities ; 3 <i>idea of</i> outside air has not been excluded so change in colour may not be due only to oxygen produced ; 4 <i>idea that</i> colour change is, subjective / difficult to see ; 5 <i>idea that</i> only, one species / a protocist / a unicell, is tested (so may not be valid for all species / types, of photosynthetic organisms ;	2